



Data Analytics Capstone

Is the Remote World the Future?

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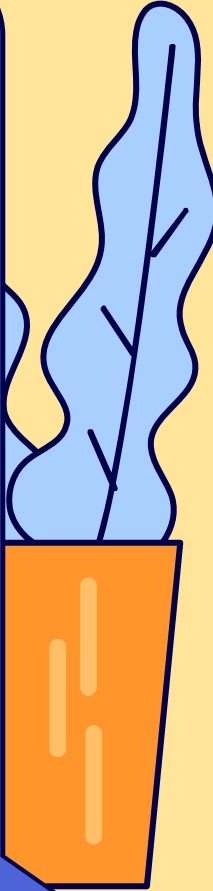


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01

INTRODUCTION

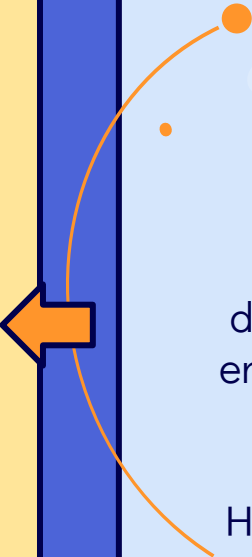
This analysis was conducted to gain insight to the impact of remote work on job performance and overall employee satisfaction as more companies transition to in-person and on site requirements post covid and quarantine years.



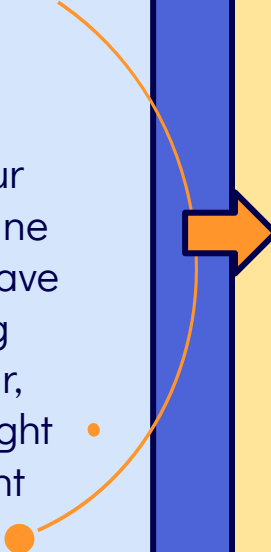


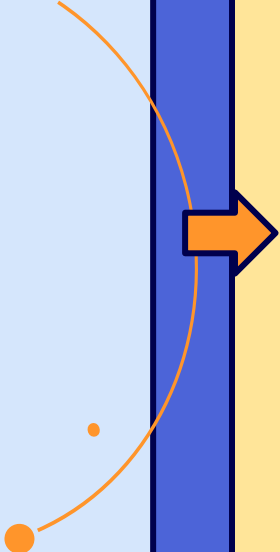
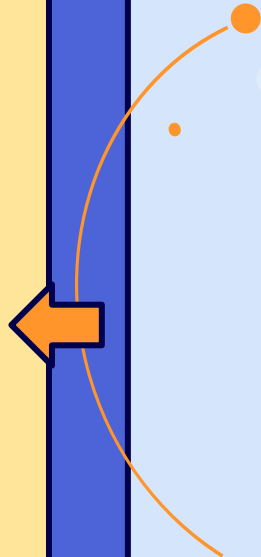
01

MOTIVATION



The post Covid and Quarantine era catapulted our society, especially the corporate world, to be heavily technology oriented. From how we submit our deliverable to how we meet our teams. I started at healthfirst in the quarantine era in a remote position, so my experience is a lot different than those who have always worked inside of the office. Recently, various companies, including Healthfirst, are trending to hybrid or fully in-person structure. Early last year, Healthfirst collected data on remote work from the employees to gain an insight to the impact and preference that team members and leaders hold. I sought data that reflected a similar scope to explore how the original data was analyzed or how companies can utilize this data in the future.





02

DATA SOURCE

Kaggle: Employee Performance and Productivity Data

<https://www.kaggle.com/datasets/mexwell/employee-performance-and-productivity-data>

Data Card: 20 Columns 100,000 Rows

Synthetic data generated by GPT (Dataset creator used OpenAI's models to create and artificial dataset that mimic the statistical properties and structures of real-world data)



03

METHODOLOGIES

Downloaded dataset and analyzed information using Python in Google Colab

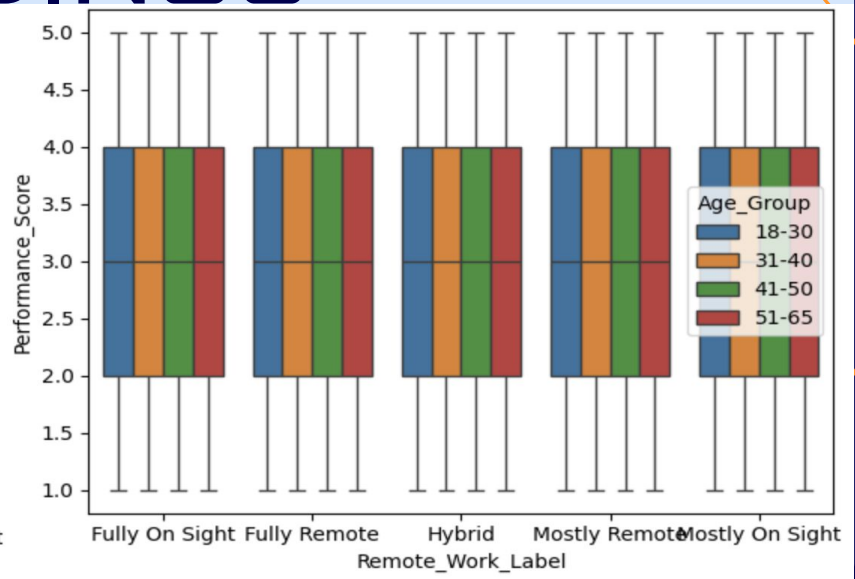
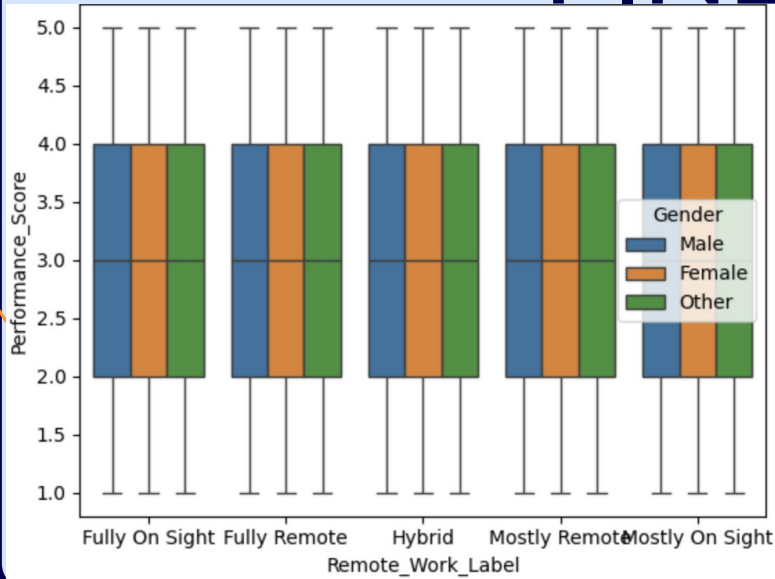
Imported Libraries: pandas; seaborn and matplotlib.pyplot to clean, analyze, group, transform and visualize the data

Research Questions



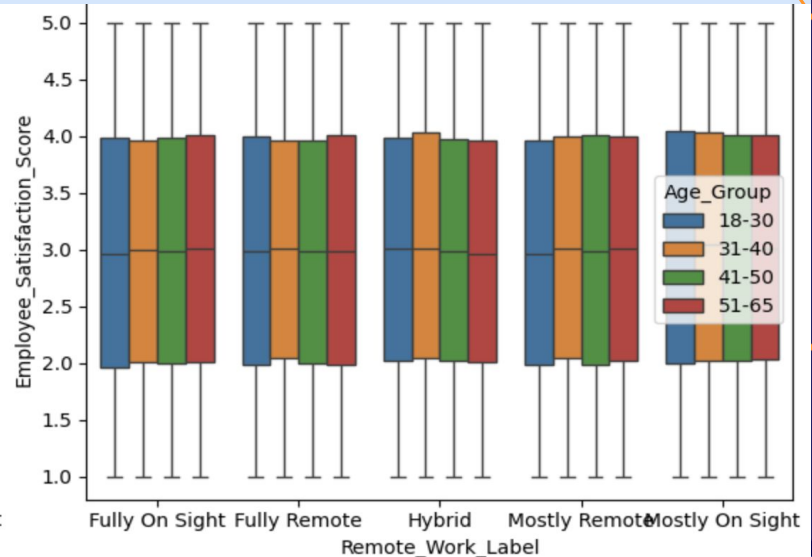
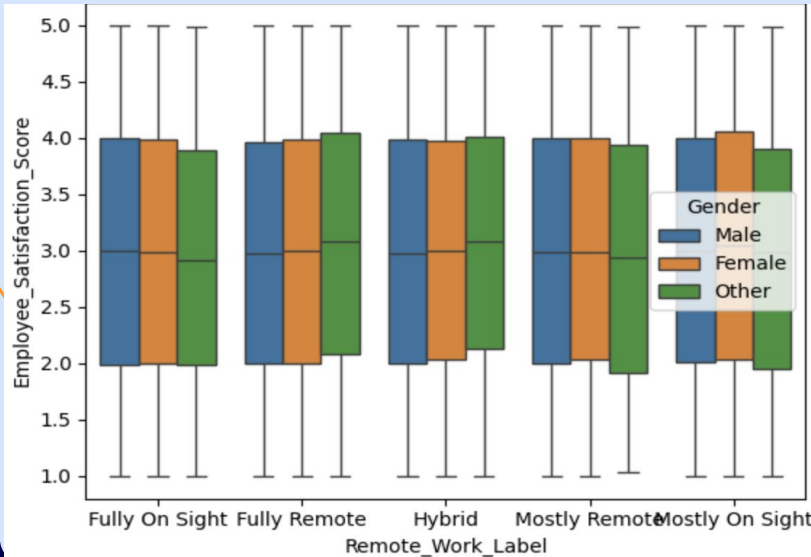
- Does Remote Work Affect Overall Employee Job Satisfaction?
- Does Remote Work Affect Employee Performance?
- Are there other factors that impact Employee Job Satisfaction and Performance more than Remote impact?
- Should companies continue to adjust toward a remote/ hybrid work environment?

04 FINDINGS



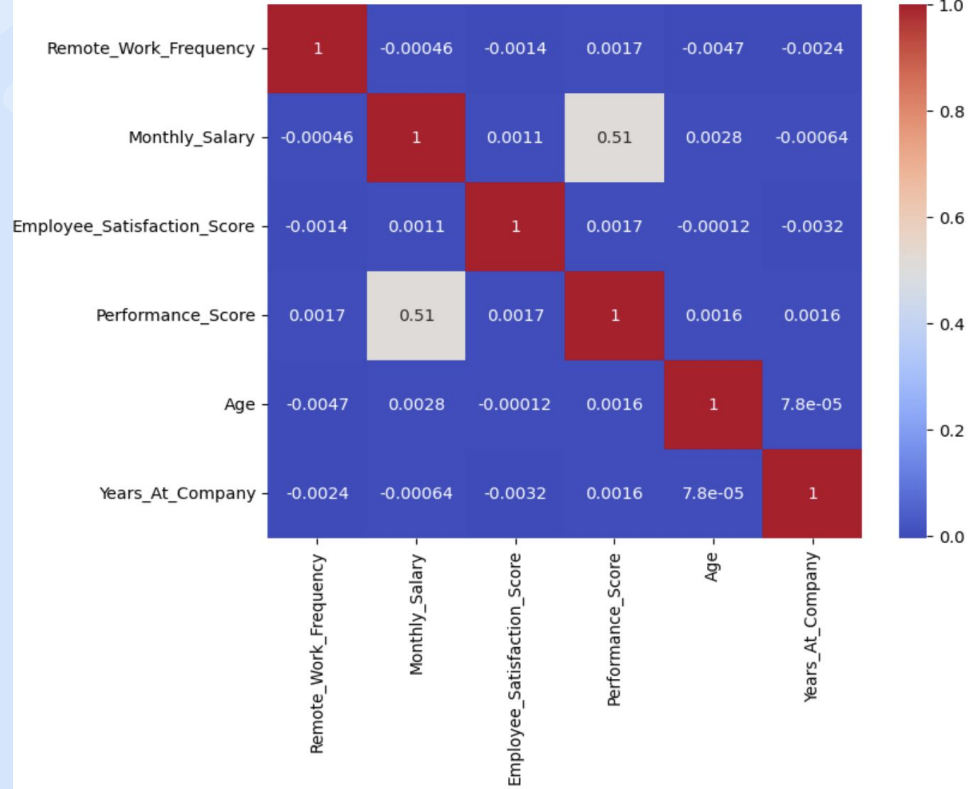
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FINDINGS cont'





Correlation Between Workplace Factors



Summary

Performance Score

Averages out with no demonstration of impact from in person or remote work

Satisfaction Score

Averages out with no demonstration of impact from in person or remote work

Age

Though only slight, Age demonstrated more of a relationship on satisfaction score than remote work in general. No impact on performance

Gender

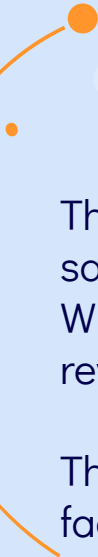
Though only slight, Gender demonstrated more of a relationship on satisfaction score than remote work in general. No impact on performance

Age and gender did not demonstrate significant impact. The slight variation demonstrates ages 31-40 may have a slight higher job satisfaction in hybrid positions but unable to develop a firm conclusion that job location is the deciding factor. As for gender variations people that did not align male or female but fell into other demonstrated a slight impact on job satisfaction by remote and hybrid work vs on site option. Potential social implications and effects of the workplace.



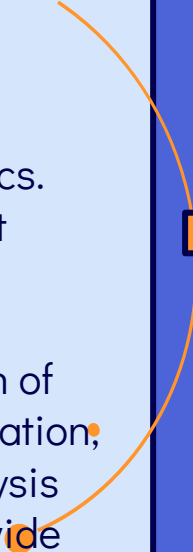
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Conclusion



The project examined whether remote work meaningfully affects employee satisfaction, employee performance, promotions, and workforce demographics. While some variation existed across remote work groups, the analysis did not reveal substantial differences or strong correlations.

This suggests that employee outcomes are likely influenced by a combination of factors beyond remote work availability, including but not limited to compensation, education, organizational structure, and job-specific conditions. Future analysis with additional workplace context or real-world organizational data may provide deeper insight into these relationships.



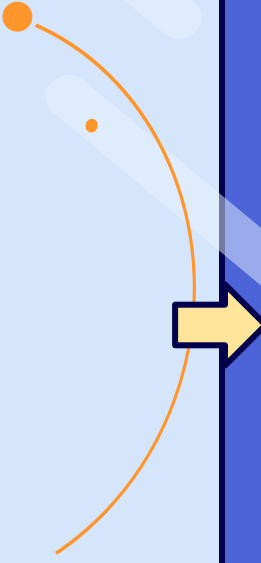
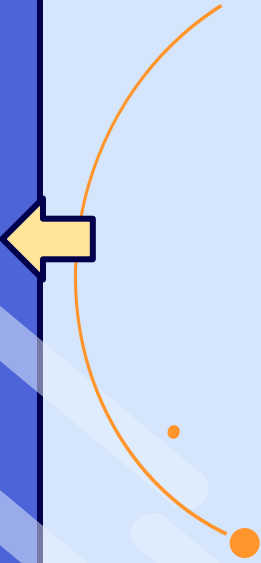
Acknowledging Gaps

It is important to recognize factors that may influence my findings:

Synthetic Data
Remote Policy Context
Industry / Role Differences
Individual Employee circumstances
Satisfaction/ Performance scores are Subjective
Correlation $\neq$ Causation

**Looking at a more aggregate level to measure the impact of remote work on economic performance across 43 private sector industries, Fernald et al. (2024) find little relationship between labor productivity and the ability of workers in an industry to work entirely remotely, suggesting remote work neither hindered nor helped raise aggregate productivity growth. (BLS)

06



“The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking.”

—Albert Einstein



THANK YOU!

Questions?

Comments?

